

Prime-power structure of the stable regime for modular Schur numbers

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Abstract

For a modulus m and a sufficiently large sum length ℓ , put $d = \gcd(m, \ell - 1)$ and $n = m/d$. We prove that every nonempty safe subset of $[1, n - 1]$ lies inside a canonical congruence block determined by one prime divisor of n , and that every such block is safe. The stable modular Schur problem therefore becomes an exact finite set-cover problem. We split this cover into mandatory support-one seeds and a residual cover, count the seeds in closed form, and show how multiplication by a prime transports the residual problem. Iteration removes every exponent beyond a canonical critical core and gives an explicit additive contribution for the removed prime-power layers. We also give exact prefix formulas for a prime-power core and a forced-block-plus-matching formula when exactly two active primes remain. The paper separates Lean-formalized results, audited prose proofs, and the open residual problem on critical cores with three or more axes.

1. Introduction

The modular Schur number $S_m(k, \ell)$, introduced in [CMD13], is the largest N for which $1, \dots, N$ can be colored with k colors and no color contains

$$x_1 + \dots + x_\ell \equiv y \pmod{m}.$$

The endpoint formula established in the companion draft, *Endpoint formulas for modular Schur numbers, with a correction to a prime-power formula*, uses

$$d = \gcd(m, \ell - 1), \quad n = m/d$$

and shows that $S_m(k, \ell) = n - 1$ once enough colors are available. The present paper asks the next question: how many colors are actually necessary?

These questions belong to Ramsey theory on the integers and to additive combinatorics; standard background appears in [BB82, LR14]. The later formulas,

tables, and open problems of D’orville, Sim, Wong, and Ho [DSWH25] provide the immediate literature context. The structural prime-block results below are contributions of the present project.

For large ℓ , this question has a rigid prime-power structure. Every safe color class fits inside a congruence block attached to one prime divisor of n . A coloring is therefore a cover of $[1, n - 1]$ by these blocks. Some blocks are mandatory because they contain a point found in no other block. After deleting them, one smaller residual cover remains.

The main results are:

1. a canonical-block description of every safe set in the stable range;
2. an exact seed-plus-residual decomposition and a closed seed count;
3. an exact recurrence when a prime exponent is extended;
4. reduction to an exponent-truncated **critical core**;
5. exact formulas for prime-power prefixes and cores with at most one active prime; and
6. an exact bipartite-matching formula when two active primes remain.

No advanced algebra is assumed. We introduce valuations, cyclic subgroups, set covers, and matchings as they appear. The main unresolved problem is also stated precisely: after the forced prime-power tails are removed, determine the residual cover on a critical core with three or more active prime axes.

2. Stable lengths and a coset lemma

A subset C of the residues modulo m is ℓ -**sum-free** if it contains no $x_1, \dots, x_\ell, y$ with $x_1 + \dots + x_\ell \equiv y \pmod{m}$. We work in the stable range

$$\ell \geq \max(2, m - 1).$$

For a prime p and positive integer r , the p -**adic valuation** $v_p(r)$ is the exponent of p in the prime factorization of r . This is only a counting device here; no analysis over the p -adic numbers is used.

Choose $a \in C$ and let

$$H = \langle C - C \rangle$$

be the smallest subgroup of $\mathbb{Z}/m\mathbb{Z}$ containing all differences $c - c'$. It has the form $g\mathbb{Z}/m\mathbb{Z}$ for a divisor g of m .

Lemma 2.1 (coset saturation). If $|C| \geq 2$ and $t \geq |H| - 1$, then

$$tC = ta + H.$$

Consequently C is unsafe at length t precisely when $g \mid (t-1)a$.

Proof. Put $D = C - a$. Then $0 \in D$ and D generates H . Because $0 \in D$, the chain $D \subseteq 2D \subseteq 3D \subseteq \dots$ is increasing. If a step fails to grow, say $jD + D = jD$, then translation by each member of D preserves jD . Hence translation by every member of the subgroup generated by D preserves it, forcing $jD = H$. Before that happens the size increases at every step, so it happens by step $|H| - 1$. Translating back gives $tC = ta + H$.

All of C lies in $a + H$. The cosets $a + H$ and $ta + H$ meet precisely when $(t-1)a \in H$, which says $g \mid (t-1)a$. $\square$

Theorem 2.2 (stable dependence on the gcd). Let $\ell, \ell' \geq \max(2, m-1)$ and suppose

$$\gcd(m, \ell - 1) = \gcd(m, \ell' - 1) = d.$$

Every subset of residues has the same safety status at lengths ℓ and ℓ' . Therefore $S_m(k, \ell) = S_m(k, \ell')$ for every $k \geq 1$.

Proof. Write $m = dn$, $\ell - 1 = du$, and $\ell' - 1 = du'$. Then both u and u' are coprime to n . A singleton $\{x\}$ is unsafe at either length precisely when $n \mid x$.

For $|C| \geq 2$, use Lemma 2.1 and write $H = g\mathbb{Z}/m\mathbb{Z}$. Put $e = \gcd(g, d)$ and $h = g/e$. From $g \mid dn$ one obtains $h \mid n$. Also $g \mid dux$ holds precisely when $h \mid x$, because h is coprime to both d/e and u . The same calculation with u' gives the same test. Thus each class, each partition, and each feasible prefix has the same status at both lengths. $\square$

We may therefore write $S_{m,d}^\infty(k)$ for the common stable value.

3. Canonical prime blocks

Assume $n = m/d \geq 2$ and put

$$U_n = [1, n-1].$$

For every prime p , define

$$a_p = v_p(d), \quad b_p = v_p(n).$$

Thus $b_p = 0$ when $p \nmid n$. Canonical blocks themselves use only primes dividing n .

If $x \in U_n$ and $t = v_p(x) < b_p$, define the canonical block

$$B_{p,x} = \{y \in U_n : y \equiv x \pmod{p^{a_p+t+1}}\}.$$

Different anchors can name the same subset. When counting blocks, duplicate subsets are counted once unless an axis label is explicitly retained.

Theorem 3.1 (canonical-block normal form).

1. Every canonical block is ℓ -sum-free.
2. Every nonempty ℓ -sum-free set $C \subseteq U_n$ is contained in a canonical block.

Proof. Fix $B_{p,x}$ and put $t = v_p(x)$. Every difference of two members of the block is divisible by p^{a_p+t+1} . If the subgroup generated by the differences is $g\mathbb{Z}/m\mathbb{Z}$, then $v_p(g) \geq a_p + t + 1$. After removing the common factor with d , the reduced divisor $h = g/\gcd(g, d)$ has $v_p(h) \geq t + 1$, so $h \nmid x$. The divisibility test in the proof of Theorem 2.2 and Lemma 2.1 show that the block is safe. The singleton case follows directly from $x < n$.

Conversely, take a nonempty safe C and $x \in C$. For a singleton, choose a prime $p \mid n$ for which $v_p(x) < b_p$; one exists because $n \nmid x$. Then $C \subseteq B_{p,x}$.

If $|C| \geq 2$, write $\langle C - C \rangle = g\mathbb{Z}/m\mathbb{Z}$ and $h = g/\gcd(g, d)$. Safety and Lemma 2.1 imply $h \nmid x$, while the calculation in Theorem 2.2 gives $h \mid n$. Choose p with $v_p(h) > v_p(x) = t$. Then $v_p(g) \geq a_p + t + 1$. Every difference $y - x$, with $y \in C$, is divisible by g and therefore by p^{a_p+t+1} . Hence every y lies in $B_{p,x}$. $\square$

For $0 \leq N \leq n - 1$, let $\kappa_{m,d}(N)$ be the smallest number of distinct canonical blocks whose union contains $[1, N]$.

Corollary 3.2 (exact prefix reduction). The smallest number of safe classes needed to partition $[1, N]$ is $\kappa_{m,d}(N)$. Consequently

$$S_{m,d}^\infty(k) = \max\{0 \leq N \leq n - 1 : \kappa_{m,d}(N) \leq k\}.$$

Proof. A safe partition can be enlarged class by class to a cover by canonical blocks using Theorem 3.1, so the cover number is no larger. In the other direction, assign each point to one selected block containing it, breaking ties arbitrarily. The assigned subsets partition the prefix and are safe because they lie inside safe blocks. $\square$

Write $\kappa(n, \mathbf{a}) = \kappa_{m,d}(n - 1)$ when only the quotient n and the depth data $\mathbf{a} = (a_p)$ matter.

4. Mandatory seeds and the residual cover

We first give a general set-cover lemma. Let L be a finite set of labels and let $N(i)$ be the set of points covered by label i . Two labels may have the same neighborhood. A label is a **seed** if it has a private point: some $x \in N(i)$ belongs to no $N(j)$ with $j \neq i$.

Let F be the set of seed labels. Delete every point covered by a seed and restrict every nonseed neighborhood to the remaining points. Let τ_{res} be the minimum size of a cover of this residual universe.

Lemma 4.1 (seed deletion). The original cover number is

$$\tau = |F| + \tau_{\text{res}}.$$

Proof. Every cover must select each member of F , because its private point has no other covering label. After these mandatory labels are chosen, exactly the residual universe remains. Adding a minimum residual cover gives a cover of size $|F| + \tau_{\text{res}}$, while restricting any original cover after its forced labels are removed gives the opposite inequality. Private points also show that two seeds cannot have identical neighborhoods, so no duplicate-set count is hidden in $|F|$. $\square$

For canonical blocks, a point x lies on the p -axis when $v_p(x) < b_p$. Its **support** is the set of all such prime axes. The seeds are exactly the blocks having a point whose support consists of one prime.

Theorem 4.2 (closed seed count). Define

$$K_{\text{seed}}(n, \mathbf{a}) = \sum_{p|n} (p-1) \sum_{j=0}^{b_p-1} p^{\min(a_p, j)}.$$

Then the number of canonical seed blocks is $K_{\text{seed}}(n, \mathbf{a})$, and

$$\boxed{\kappa(n, \mathbf{a}) = K_{\text{seed}}(n, \mathbf{a}) + \tau_{\text{res}}(n, \mathbf{a}).}$$

Proof. A support-one witness on the p -axis is divisible by q^{b_q} for every other prime $q \mid n$. Put $M = n/p^{b_p}$. Every such witness at layer t has the unique form

$$x = Mp^t u, \quad 1 \leq u < p^{b_p-t}, \quad p \nmid u.$$

After division by p^t , its block is determined by Mu modulo p^{a_p+1} . Since M is coprime to p , multiplication by M permutes the unit residues. The range contains $(p-1)p^{b_p-t-1}$ possible units, while a complete unit residue system modulo p^{a_p+1} has $(p-1)p^{a_p}$ members. There is no repetition before the smaller of these two counts is reached. Thus the number of represented blocks is

$$(p-1)p^{\min(a_p, b_p-t-1)}$$

such residues. Blocks at different layers or on different axes have different private witnesses. Summing over $t = 0, \dots, b_p-1$ and reversing the index by $j = b_p-t-1$ gives the displayed inner sum. Lemma 4.1 then gives the cover decomposition. $\square$

5. Multiplying by a prime

Let p be prime, let p^{b_p} be the exact power of p dividing n , and assume

$$a_p \leq b_p.$$

This is the **stable-depth** condition. Let C_n be the points covered by seeds and $R_n = U_n \setminus C_n$ the residual points.

Theorem 5.1 (residual transport). Under the stable-depth condition,

$$C_{pn} = \{y \in U_{pn} : p \nmid y\} \sqcup \{px : x \in C_n\},$$

and hence

$$\boxed{R_{pn} = \{px : x \in R_n\}.}$$

Moreover

$$\tau_{\text{res}}(pn, \mathbf{a}) = \tau_{\text{res}}(n, \mathbf{a}).$$

Proof. First suppose $p \mid n$. Multiplication by p preserves the full list of supported prime axes: on the p -axis both the point valuation and the ambient exponent rise by one; on every other axis the valuation is unchanged. Thus px has a support-one witness precisely when x does.

If $p \nmid n$, then $b_p = 0$, so the depth assumption forces $a_p = 0$. The old universe has no p -axis. After multiplication by p , each transported old point has p -valuation at least the new ambient exponent and remains unsupported on that axis. Its old support is again unchanged.

Every new point not divisible by p lies at p -valuation zero. The inequality $a_p \leq b_p$ ensures that every represented valuation-zero p -block has a support-one witness, so all such points are seed-covered. This proves the two point-set identities.

Write a canonical label as (q, t, r) : its prime axis, valuation layer, and residue. Define

$$\Psi(q, t, r) = \begin{cases} (p, t+1, pr), & q = p, \\ (q, t, pr \bmod q^{a_q+t+1}), & q \neq p. \end{cases}$$

The first branch is absent when p is fresh. The residual neighborhood of $\Psi(q, t, r)$ is exactly the pointwise multiple by p of the old residual neighborhood. Conversely, every **nonempty** new residual neighborhood has an old producer: choose a point px in it and take the old label through x on the same axis.

Send an old minimum cover forward and discard duplicate extensional neighborhoods; its size cannot increase. In the reverse direction, discard selected labels with empty neighborhoods, choose one old producer for each remaining neighborhood, and again discard duplicates. Its size cannot increase either. The two minimum cover sizes are equal. $\square$

The seed count increases by one full stabilized layer:

$$K_{\text{seed}}(pn, \mathbf{a}) = K_{\text{seed}}(n, \mathbf{a}) + (p-1)p^{a_p}.$$

Combining this with Theorems 4.2 and 5.1 gives the main recurrence.

Corollary 5.2 (one-step recurrence).

$$\boxed{\kappa(pn, \mathbf{a}) = \kappa(n, \mathbf{a}) + (p-1)p^{a_p}.$$

The condition $a_p \leq b_p$ is necessary. Below that depth, new nonmultiples of p need not be seed-covered, so the residual point identity can fail.

6. Inactive primes and prime-power cores

Call $p \mid n$ **active** when $a_p > 0$ and **inactive** when $a_p = 0$. Factor

$$n = n_A n_I,$$

where n_A contains the active prime powers and n_I the inactive ones. Put

$$d_A = \prod_{p \mid n_A} p^{a_p}, \quad m_A = d_A n_A,$$

and use the convention that the terminal cover of $n_A = 1$ is 0. If $p^b \parallel n_I$, its one-axis blocks are indexed by a layer $0 \leq t < b$ and a nonzero residue modulo p , giving $b(p-1)$ mandatory blocks.

Theorem 6.1 (inactive-prime stripping). At the terminal prefix,

$$\boxed{\kappa(n, \mathbf{a}) = \sum_{p^b \parallel n_I} b(p-1) + \kappa_{m_A, d_A}(n_A - 1).$$

When $n_A = 1$, the final term is 0.

Proof. For each inactive block, choose a point divisible by every other prime power and lying in the prescribed p -layer and nonzero residue. This point lies

on only that axis, so the block is mandatory. Their union is exactly the set of points not divisible by n_I . The remaining points are

$$n_I [1, n_A - 1].$$

Division by n_I preserves incidence with every active canonical block. Thus the residual cover is $\kappa_{m_A, d_A}(n_A - 1)$, while the mandatory inactive blocks contribute the displayed sum. $\square$

For a prime-power quotient $n = p^b$, define

$$R_t(N) = \left\lfloor \frac{N}{p^t} \right\rfloor - \left\lfloor \frac{N}{p^{t+1}} \right\rfloor,$$

the number of integers at most N with p -valuation exactly t , and

$$F_{p,a,b}(N) = \sum_{t=0}^{b-1} \min((p-1)p^a, R_t(N)).$$

Theorem 6.2 (prime-power prefix formula). If $n = p^b$ and $a = v_p(d)$, then for $0 \leq N \leq p^b - 1$,

$$\boxed{\kappa_{m,d}(N) = F_{p,a,b}(N).}$$

Therefore

$$S_{m,d}^\infty(k) = \max\{0 \leq N \leq p^b - 1 : F_{p,a,b}(N) \leq k\}.$$

Proof. There is only one prime axis, so every point belongs to exactly one block on that axis and every represented block must be selected. At layer t , there are $R_t(N)$ points. After division by p^t , the relevant quotients are the first $R_t(N)$ positive integers not divisible by p . Their residues run through the nonmultiples of p modulo p^{a+1} , a cycle of $(p-1)p^a$ labels with no repetition before the cycle closes. The number represented is therefore the minimum in the definition of F . Summing the disjoint layers proves the formula. $\square$

Define the stable terminal threshold by

$$k_0^\infty(m, d) = \kappa_{m,d}(n - 1).$$

At $N = p^b - 1$ this becomes

$$k_0^\infty(m, d) = (p-1) \sum_{j=0}^{b-1} p^{\min(a, j)}.$$

For $p = 2$ and $a = 0$, the prefix formula simplifies to

$$S_{m,d}^\infty(k) = \min(2^k - 1, 2^b - 1).$$

7. The exponent-truncated critical core

The recurrence from Section 5 removes every exponent layer lying beyond its depth in d . Define

$$n_{\text{crit}} = \prod_{p|n} p^{\min(b_p, a_p)},$$

and let

$$e_p = b_p - \min(b_p, a_p)$$

be the number of removed layers on the p -axis.

Theorem 7.1 (critical-core reduction).

$$\kappa(n, \mathbf{a}) = \kappa(n_{\text{crit}}, \mathbf{a}) + \sum_{p|n} e_p (p-1) p^{a_p}.$$

Proof. Fix a prime p . Once the current exponent of p is at least a_p , Corollary 5.2 says that adding one further factor of p adds exactly $(p-1)p^{a_p}$ to the cover number. Starting at exponent $\min(b_p, a_p)$ and repeating this step e_p times removes the entire excess p -tail. When $a_p = 0$, the truncated core initially omits p ; its first insertion is the fresh-prime case of Theorem 5.1 and contributes $p-1$, as the same formula requires. Do this for every prime divisor of n . Prime multiplication commutes, and the added terms depend only on the chosen prime and its fixed depth, so the contributions add to the displayed sum. $\square$

The critical core is not a guessed smaller modulus. It is the unique quotient obtained by truncating each exponent b_p at the depth a_p supplied by d . All removed layers are completely understood. The remaining term $\kappa(n_{\text{crit}}, \mathbf{a})$ contains the genuine interaction among prime axes.

8. Two active primes and bipartite matching

Assume the active pair is

$$n = p^b q^c, \quad d_A = p^a q^e, \quad m_A = d_A n,$$

for distinct primes p, q , with $a = v_p(d) > 0$ and $e = v_q(d) > 0$. Fix a prefix $[1, N]$. A point supported on only one axis forces its unique block. Let $\mathcal{F}_N$ be the distinct forced blocks and delete all points they cover.

Every remaining point lies in one surviving p -block and one surviving q -block. Make a bipartite multigraph G_N : the left vertices are surviving p -blocks, the right vertices are surviving q -blocks, and each residual point is an edge joining its two blocks. Parallel edges are allowed.

Use one labelled vertex for each distinct block on each axis. If a p -block and a q -block happen to be the same extensional subset, retain both axis labels while constructing the graph.

Let $\nu(G_N)$ be the largest number of edges with no shared endpoint.

Theorem 8.1 (two-axis matching formula).

$$\kappa_{m_A, d_A}(N) = |\mathcal{F}_N| + \nu(G_N).$$

Proof. Every forced block must be selected. The labelled-cover lemma first chooses representatives when labels have duplicate extensional neighborhoods; forced blocks on opposite axes cannot coincide because their private witnesses have different support. After their points are deleted, selecting represented canonical blocks is selecting graph vertices that touch every edge: a vertex cover. Opposite-axis labels for the same remaining subset are identified when the extensional cover cardinality is counted. Thus

$$\kappa_{m_A, d_A}(N) = |\mathcal{F}_N| + \tau(G_N),$$

where τ is the minimum vertex-cover size.

For completeness, we recall the finite bipartite matching argument. Choose a maximum matching M . Starting from unmatched left vertices, follow alternating paths: unmatched edges from left to right and matched edges from right to left. Let Z be the reached vertices. Then

$$C = (L \setminus Z) \cup (R \cap Z)$$

covers every edge. Otherwise an uncovered edge would extend an alternating path, or contradict the way a matched left vertex was reached. No unmatched

right vertex is reached, since that would give an augmenting path and a larger matching. Each edge of M has exactly one endpoint in C , so $|C| = |M|$. Every vertex cover has at least $|M|$ vertices because the edges of M are disjoint. Hence $\tau(G_N) = \nu(G_N)$, proving the formula. Parallel edges do not change either quantity. $\square$

The forced contribution is explicit. With

$$F_{r,u,h}(M) = \sum_{t=0}^{h-1} \min \left((r-1)r^u, \left\lfloor \frac{M}{r^t} \right\rfloor - \left\lfloor \frac{M}{r^{t+1}} \right\rfloor \right),$$

one has

$$|\mathcal{F}_N| = F_{p,a,b} \left(\left\lfloor \frac{N}{q^c} \right\rfloor \right) + F_{q,e,c} \left(\left\lfloor \frac{N}{p^b} \right\rfloor \right).$$

Indeed, a point is supported only on the p -axis precisely when it is divisible by q^c ; division by q^c reduces its block count to the prime-power calculation of Theorem 6.2. The other term is symmetric.

At the terminal prefix,

$$k_0^\infty(m, d) = (p-1) \sum_{j=0}^{b-1} p^{\min(a,j)} + (q-1) \sum_{j=0}^{c-1} q^{\min(e,j)} + \nu(G_{n-1}).$$

9. What remains open

Theorems 4.2 and 7.1 separate the terminal cover into three parts:

forced seed count + explicit exponent-tail contribution + critical residual cover.

The first two parts have closed formulas. One active prime is handled by Section 6, and two active primes are handled by bipartite matching in Section 8. With three or more active axes, a residual point may belong to three or more blocks. The graph becomes a hypergraph, and the bipartite matching proof no longer applies.

The central open problem is:

Determine the residual cover number on every exponent-truncated critical core with at least three active prime axes.

No universal equality between cover and packing numbers is asserted here. Such an equality is false in general, including in certified project examples. Finite computations and special-cell certificates are useful evidence but do not replace a universal theorem.

10. Proof and formalization status

The mathematical claims in this draft have the following status.

Result	Mathematical status	Lean status
Stable gcd dependence and canonical-block normal form	proved; independently audited	formalization open
Exact prefix reduction	proved; independently audited	formalization open
Abstract seed deletion	proved	proved in Lean
Arithmetic seed identification and closed seed count	proved; independently audited	proved in Lean
Residual transport and one-step recurrence	proved; independently audited	proved in Lean
Terminal inactive-prime stripping	proved; independently audited	proved in Lean
Prime-power prefix and at-most-one-active formulas	proved; independently audited	formalization open
Critical-core reduction	proved; independently audited	proved in Lean
Two-active-prime matching formula	proved; independently audited	Lean checks the abstract certificate-to- <code>AETWitness</code> bridge; uniform construction of a <code>TwoAxisCertificate</code> from two-axis support remains open
Three-or-more-axis residual formula	open	open

The project-only Lean capstones have a separate transitive-axiom audit. They are outside the public Comparator’s statement-comparison gate: the Comparator does not restate these particular claims using its independent Mathlib-only definitions, does not compare those statements to solution wrappers, and does not run its per-claim nanoda check on them. The separate audit reports the

axioms used by their actual transitive dependency closures. Passing one audit should not be described as passing the other.

11. Conclusion

The stable modular Schur problem is not an unstructured search over colorings. It is an arithmetic cover whose blocks are indexed by prime, valuation layer, and residue. Private points force a computable seed contribution. Stable prime multiplication preserves the residual cover and adds a fixed amount. Iterating that fact strips the quotient down to a canonical critical core.

This gives a sharp division of labor. Prime-power tails are solved by a closed formula. A one-axis core is explicit, and a two-axis core is a matching problem. The remaining difficulty is concentrated on the finite but genuinely multi-axis residual cover of the critical core.

References

References

- [BB82] A. Beutelspacher and W. Brestovansky, *Generalized Schur numbers*, in *Combinatorial Theory* (Schloß Rauischholzhausen, 1982), Lecture Notes in Mathematics **969**, Springer, 1982, pp. 30–38.
- [CMD13] J. Chappelon, M. P. Revuelta Marchena, and M. I. Sanz Domínguez, *Modular Schur numbers*, *Electron. J. Combin.* **20**(2) (2013), #P61. arXiv:1306.5635.
- [DSWH25] J. D’orville, K. A. Sim, K. B. Wong, and C. K. Ho, *Modular generalizations of Schur numbers*, *Integers* **25** (2025), #A62.
- [LR14] B. M. Landman and A. Robertson, *Ramsey Theory on the Integers*, 2nd ed., Student Mathematical Library **73**, American Mathematical Society, 2014.